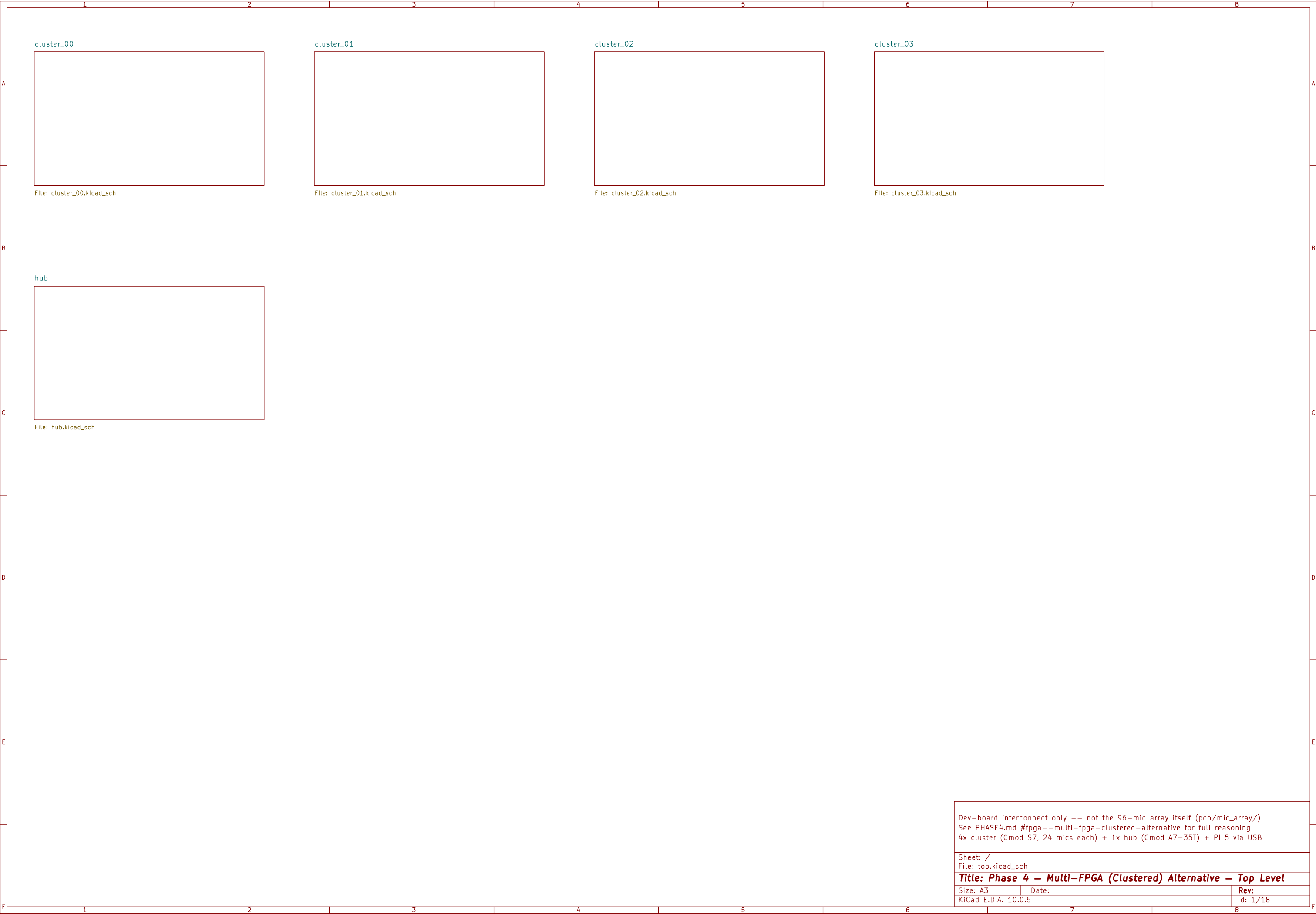




arm_00



File: arm_00.kicad_sch

arm_01

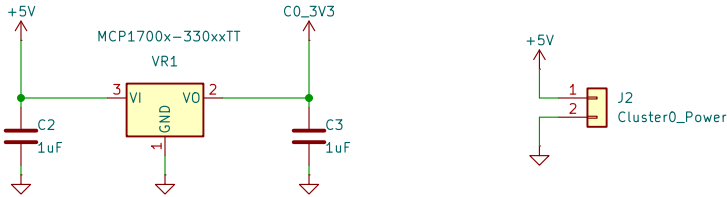
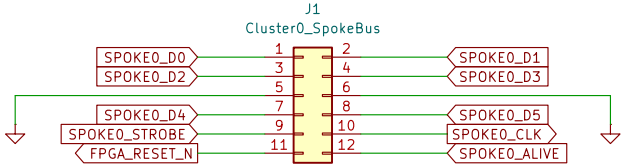
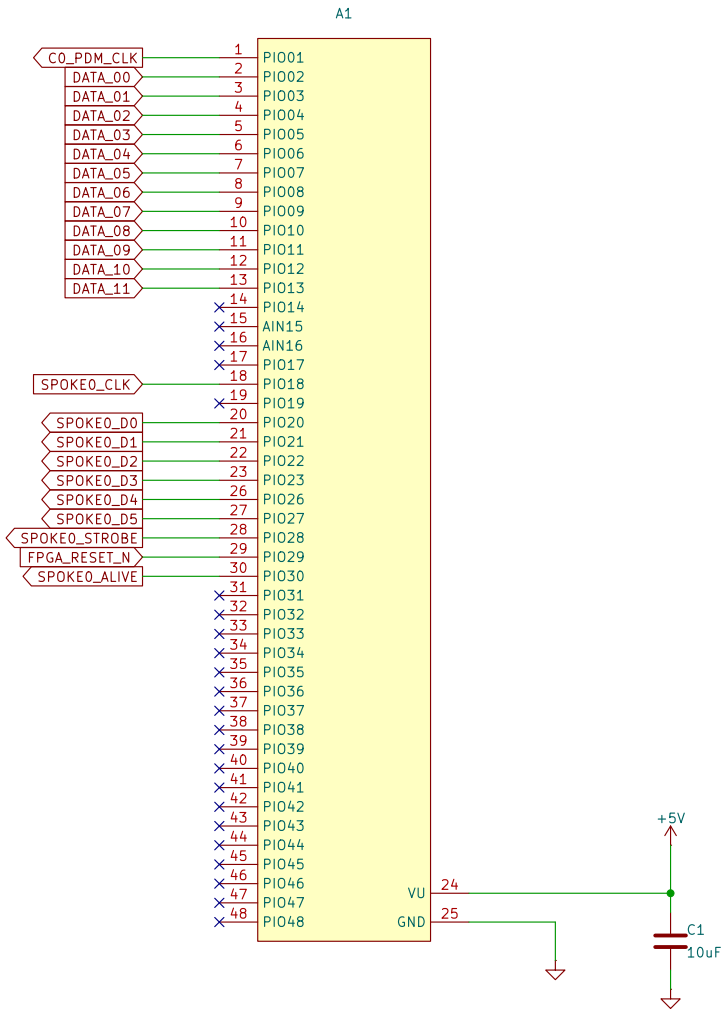


File: arm_01.kicad_sch

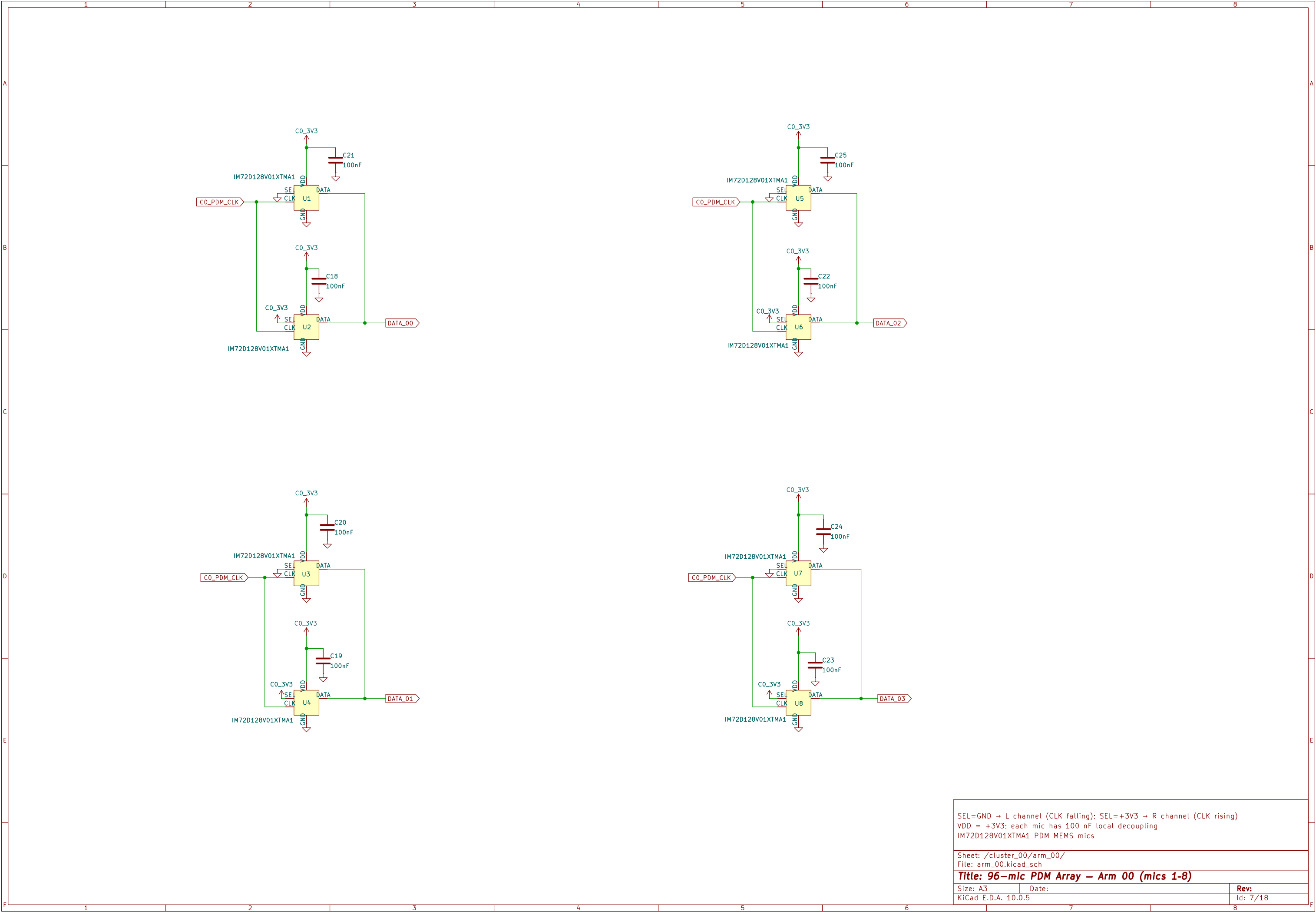
arm_02



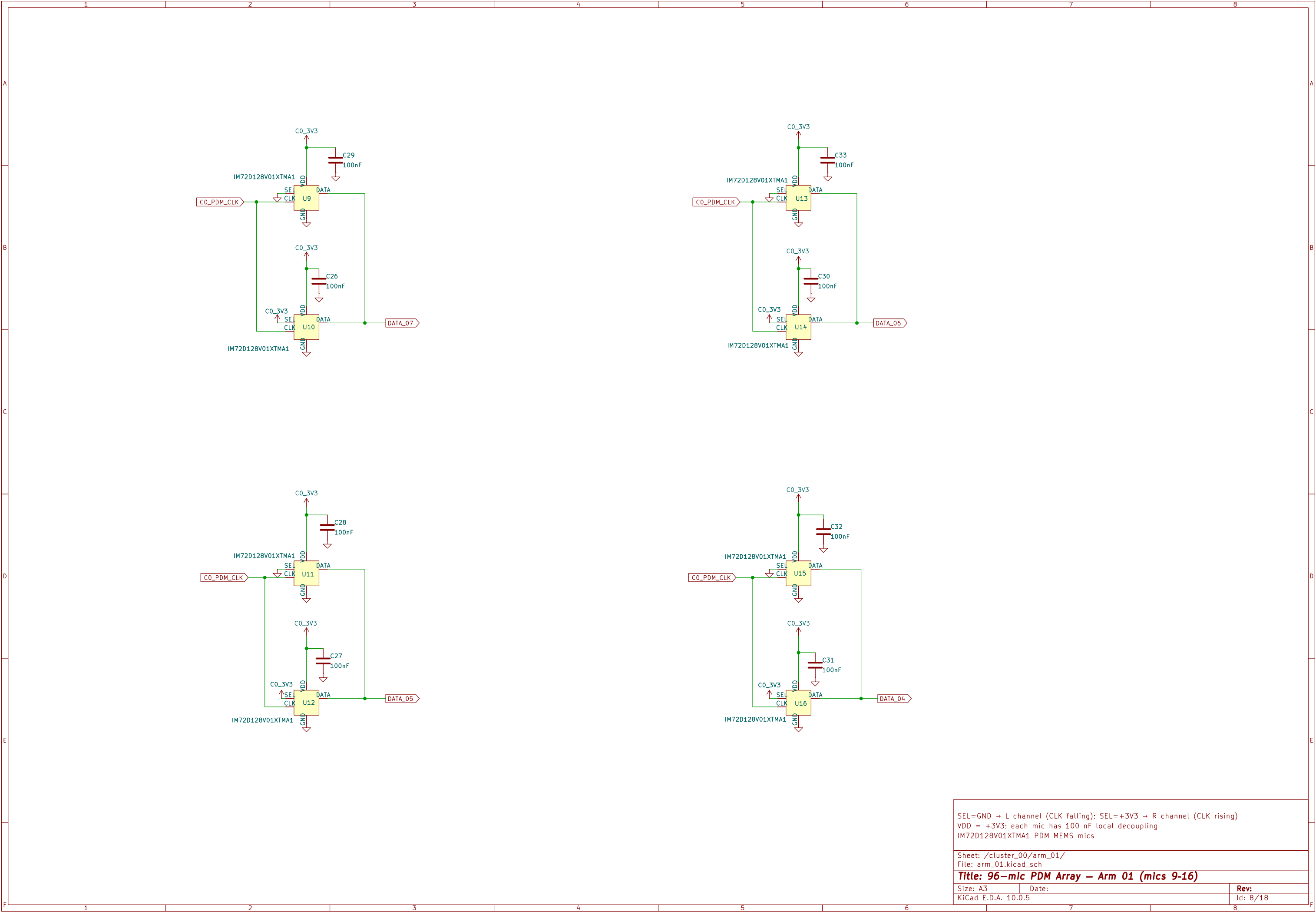
File: arm_02.kicad_sch



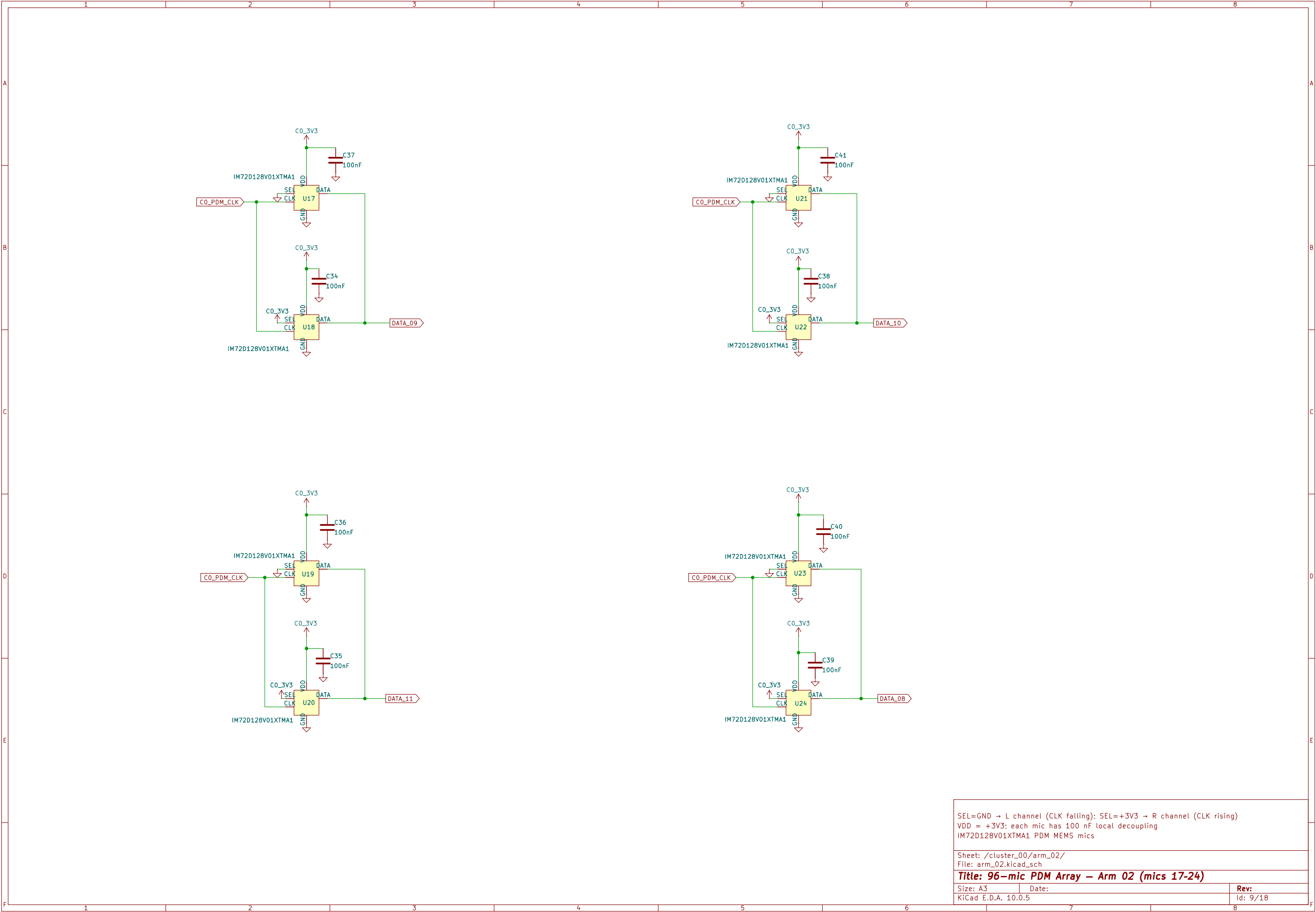
PDM: 12 data lines (2 mics/line) + 1 clock, to this cluster's 3 arms
Spoke bus: 6 data + strobe + fwd clock, parallel single-ended (not LVDS)
Digilent Cmod S7 (XC7S25) – CIC/FIR done in HDL, not shown here



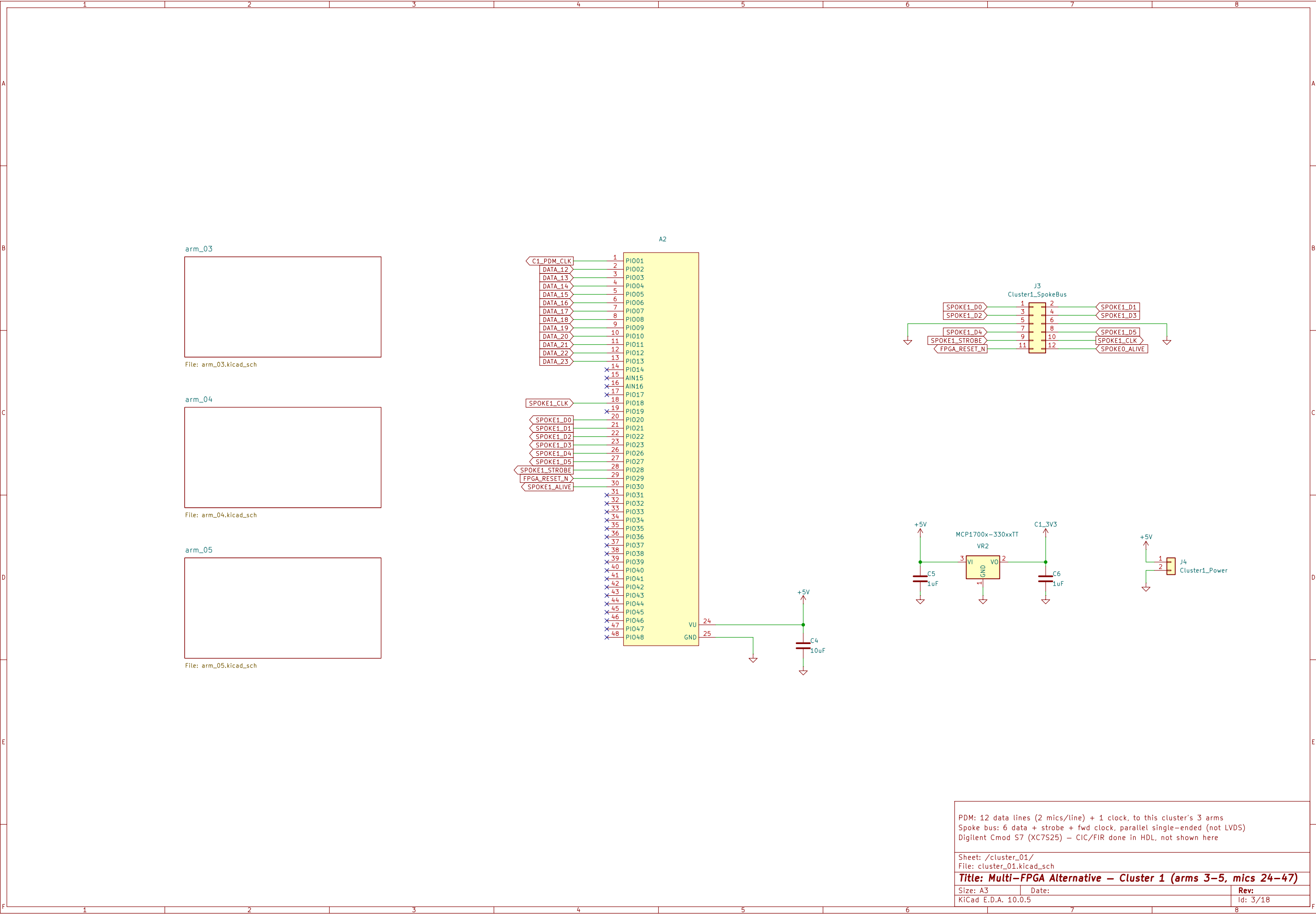
SEL=GND → L channel (CLK falling); SEL=+3V3 → R channel (CLK rising) VDD = +3V3; each mic has 100 nF local decoupling IM72D128V01XTMA1 PDM MEMS mics		
Sheet: /cluster_00/arm_00/ File: arm_00.kicad_sch		
Title: 96-mic PDM Array – Arm 00 (mics 1-8)		
Size: A3	Date:	Rev:
KiCad E.D.A. 10.0.5		Id: 7/18

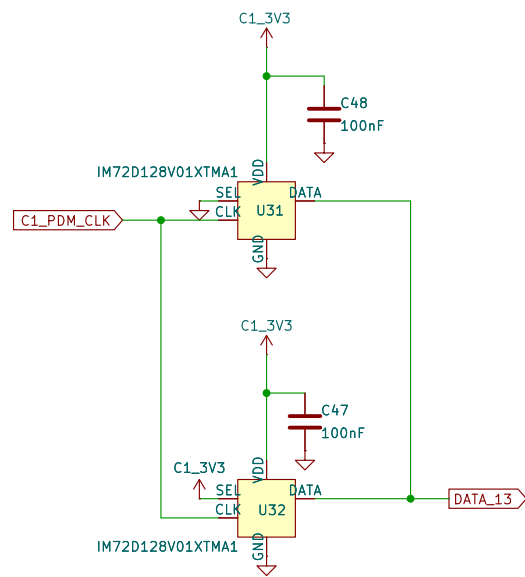
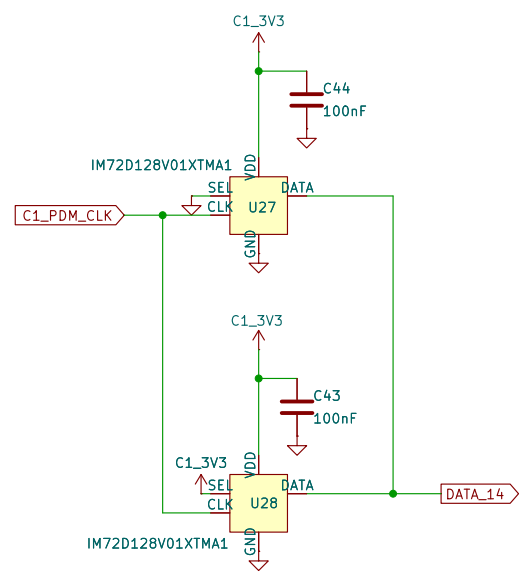
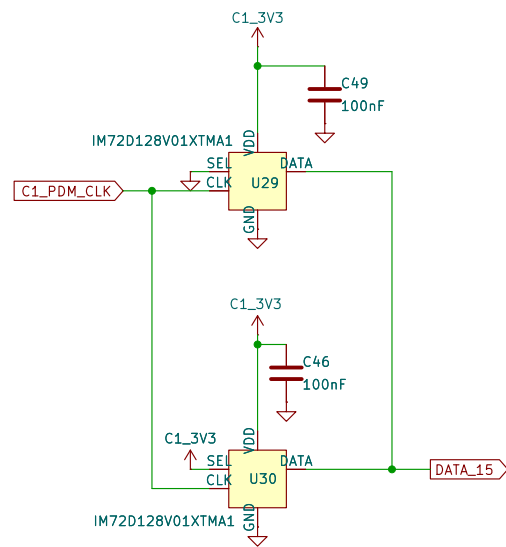
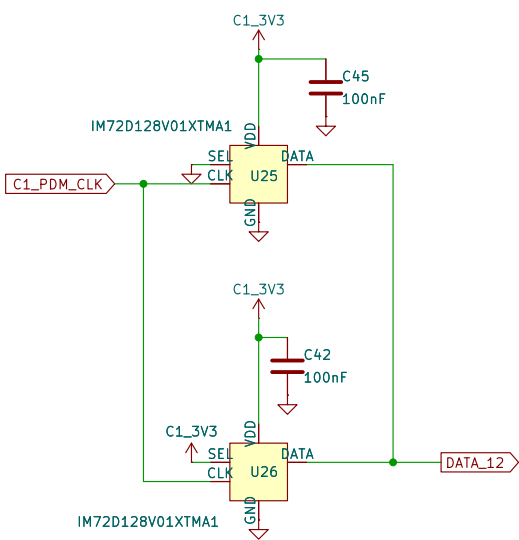


SEL=GND → L channel (CLK falling); SEL=+3V3 → R channel (CLK rising) VDD = +3V3; each mic has 100 nF local decoupling IM72D128V01XTMA1 PDM MEMS mics		
Sheet: /cluster_00/arm_01/ File: arm_01.kicad_sch		
Title: 96-mic PDM Array – Arm 01 (mics 9-16)		
Size: A3	Date:	Rev:
KiCad E.D.A. 10.0.5		Id: 8/18



SEL=GND → L channel (CLK falling); SEL=+3V3 → R channel (CLK rising) VDD = +3V3; each mic has 100 nF local decoupling IM72D128V01XTMA1 PDM MEMS mics		
Sheet: /cluster_00/arm_02/ File: arm_02.kicad_sch		
Title: 96-mic PDM Array – Arm 02 (mics 17-24)		
Size: A3	Date:	Rev:
KiCad E.D.A. 10.0.5		Id: 9/18



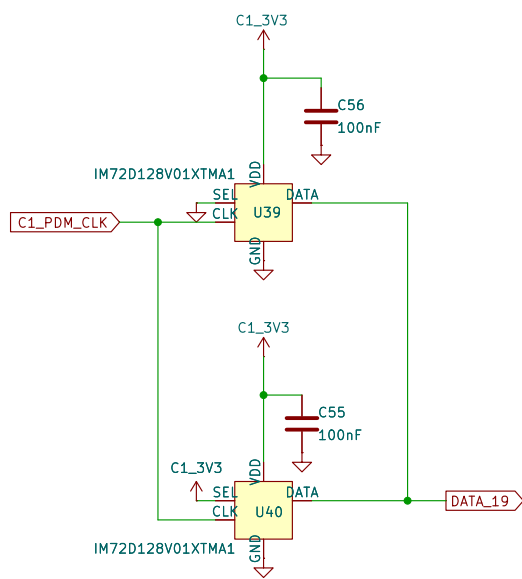
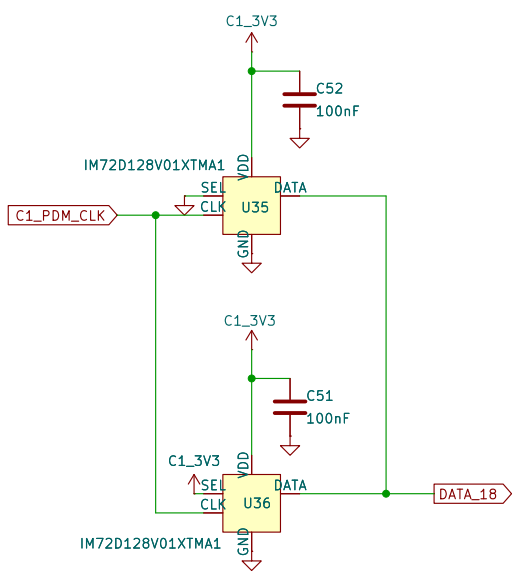
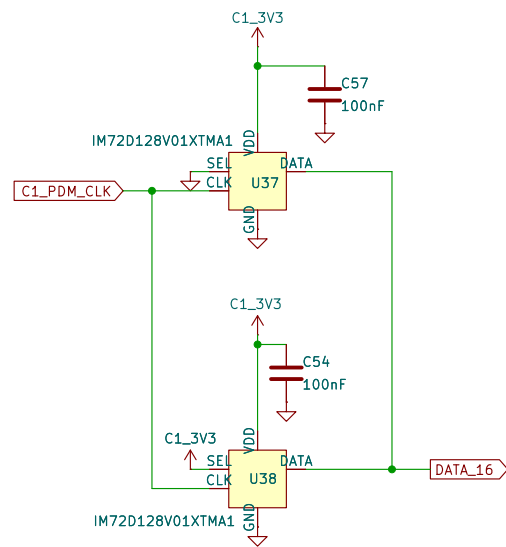
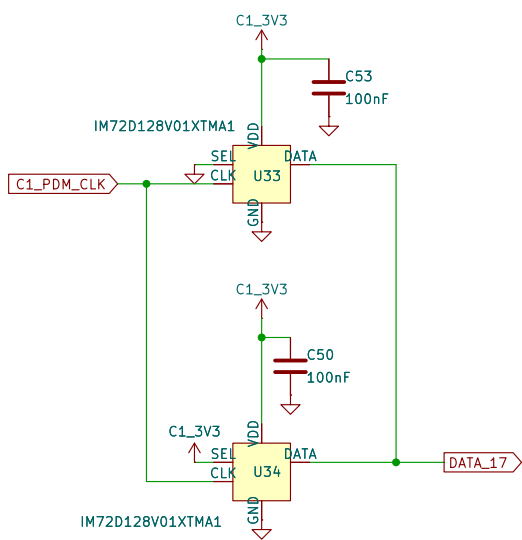


SEL=GND → L channel (CLK falling); SEL=+3V3 → R channel (CLK rising)
VDD = +3V3; each mic has 100 nF local decoupling
IM72D128V01XTMA1 PDM MEMS mics

Sheet: /cluster_01/arm_03/
File: arm_03.kicad_sch

Title: 96-mic PDM Array – Arm 03 (mics 25-32)

Size: A3	Date:	Rev:
KiCad E.D.A. 10.0.5		Id: 10/18

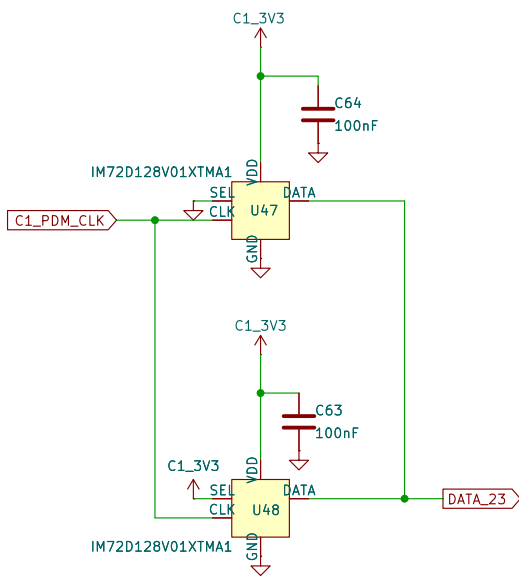
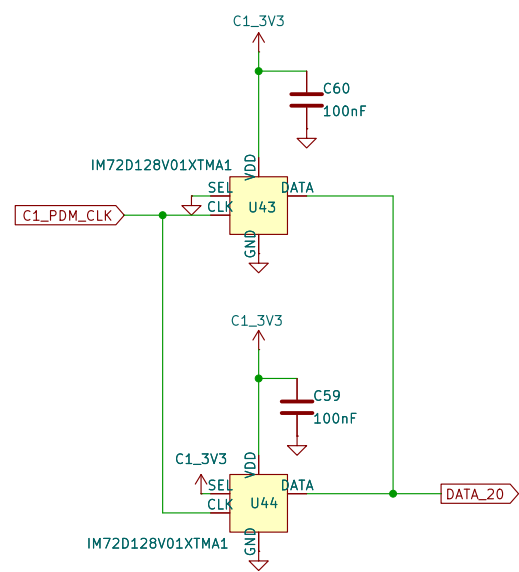
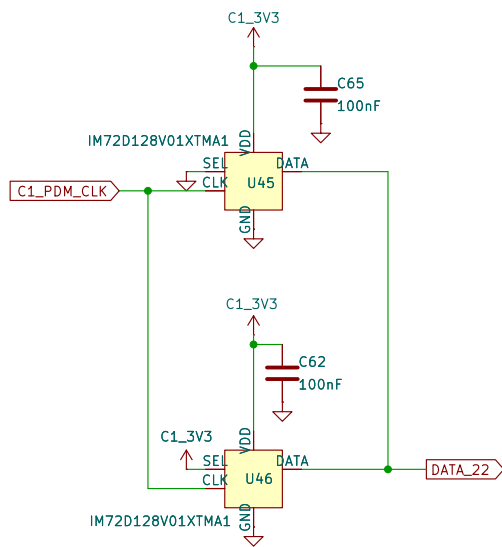
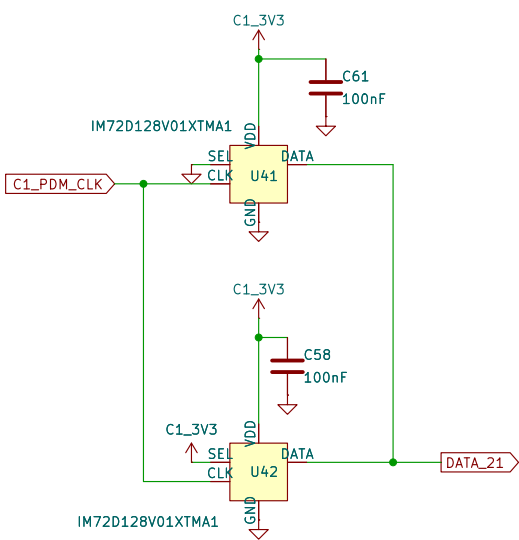


SEL=GND → L channel (CLK falling); SEL=+3V3 → R channel (CLK rising)
VDD = +3V3; each mic has 100 nF local decoupling
IM72D128V01XTMA1 PDM MEMS mics

Sheet: /cluster_01/arm_04/
File: arm_04.kicad_sch

Title: 96-mic PDM Array – Arm 04 (mics 33-40)

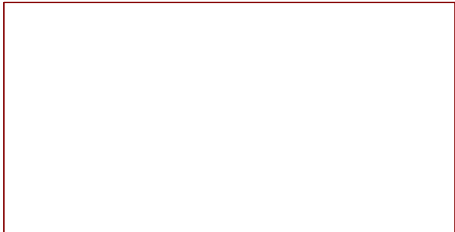
Size: A3	Date:	Rev:
KiCad E.D.A. 10.0.5		Id: 11/18



SEL=GND → L channel (CLK falling); SEL=+3V3 → R channel (CLK rising)
VDD = +3V3; each mic has 100 nF local decoupling
IM72D128V01XTMA1 PDM MEMS mics

Sheet: /cluster_01/arm_05/ File: arm_05.kicad_sch		
Title: 96-mic PDM Array – Arm 05 (mics 41-48)		
Size: A3	Date:	Rev:
KiCad E.D.A. 10.0.5		Id: 12/18

arm_06



File: arm_06.kicad_sch

arm_07

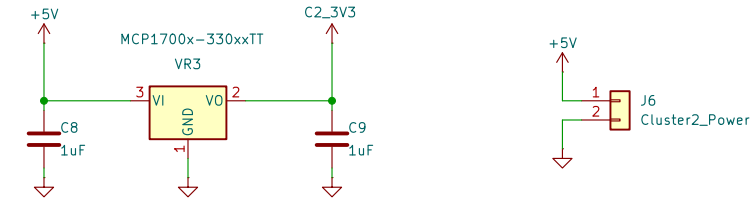
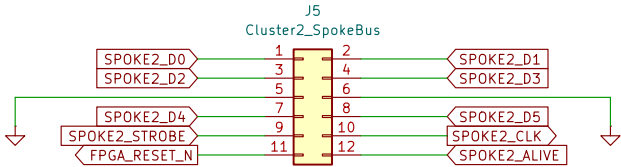
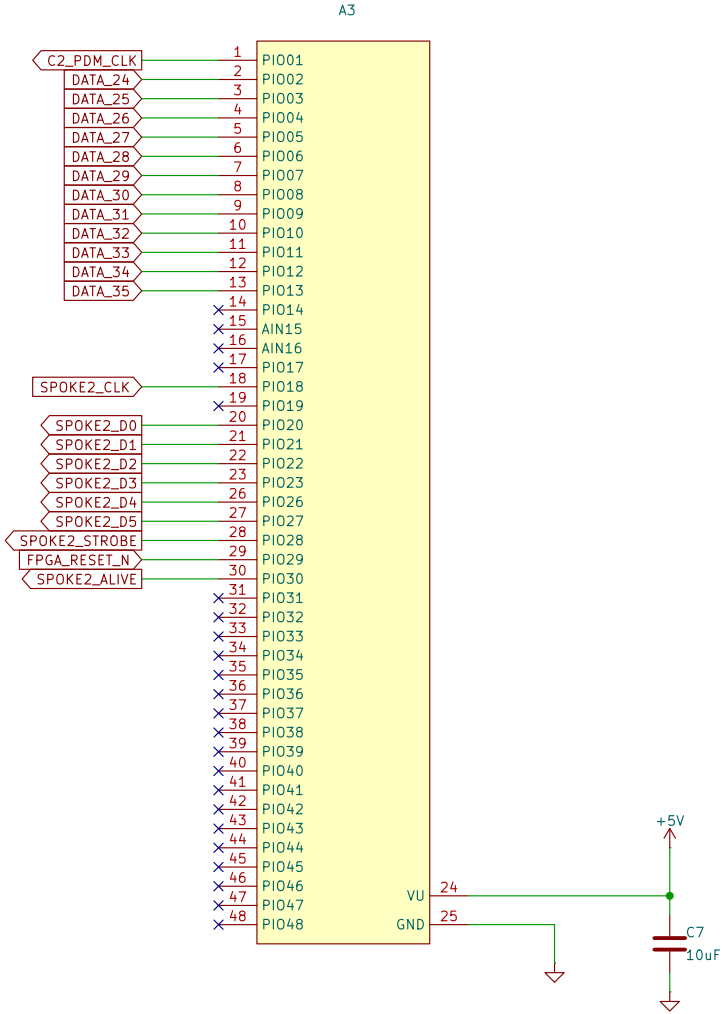


File: arm_07.kicad_sch

arm_08

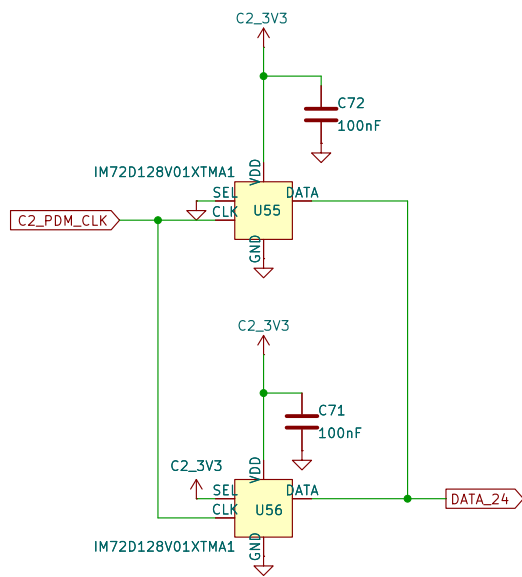
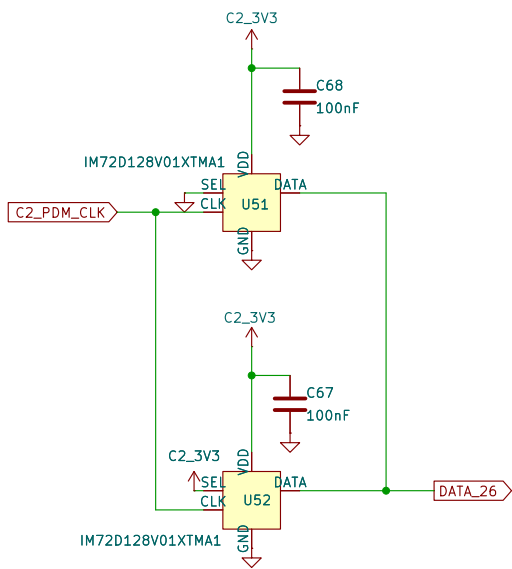
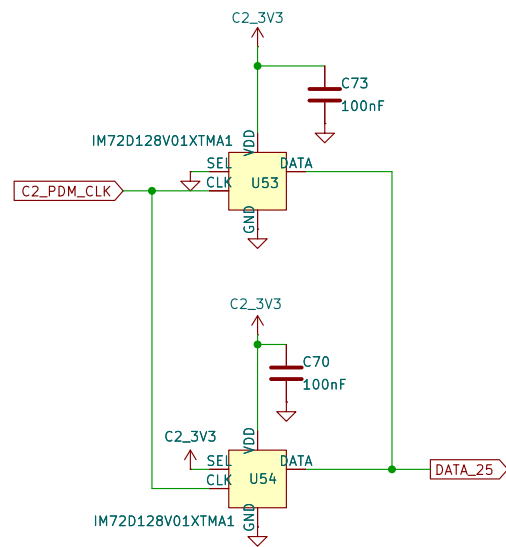
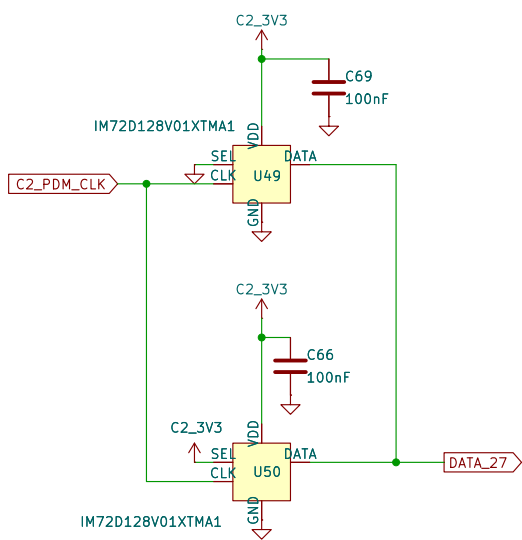


File: arm_08.kicad_sch



PDM: 12 data lines (2 mics/line) + 1 clock, to this cluster's 3 arms
Spoke bus: 6 data + strobe + fwd clock, parallel single-ended (not LVDS)
Digilent Cmod S7 (XC7S25) – CIC/FIR done in HDL, not shown here

Sheet: /cluster_02/ File: cluster_02.kicad_sch		
Title: Multi-FPGA Alternative – Cluster 2 (arms 6–8, mics 48–71)		
Size: A3	Date:	Rev:
KiCad E.D.A. 10.0.5		Id: 4/18

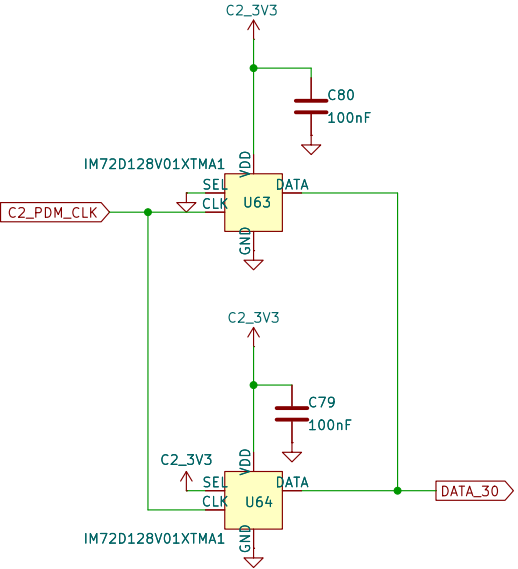
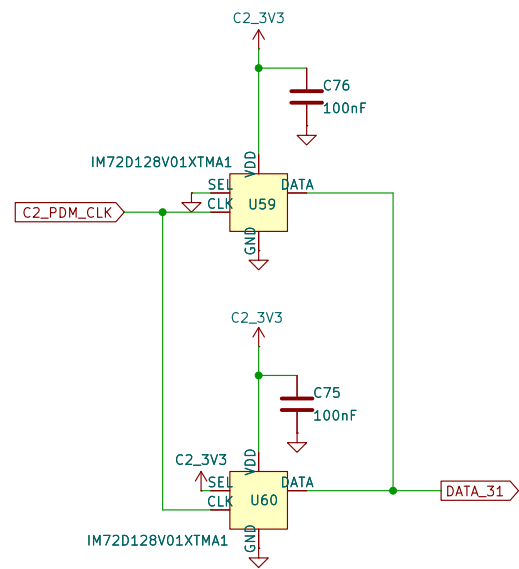
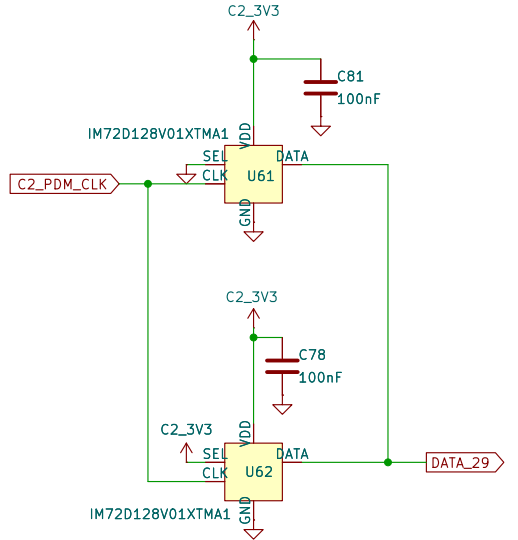
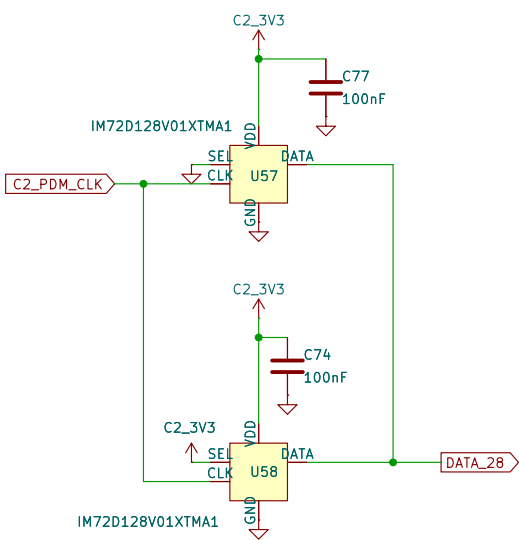


SEL=GND → L channel (CLK falling); SEL=+3V3 → R channel (CLK rising)
VDD = +3V3; each mic has 100 nF local decoupling
IM72D128V01XTMA1 PDM MEMS mics

Sheet: /cluster_02/arm_06/
File: arm_06.kicad_sch

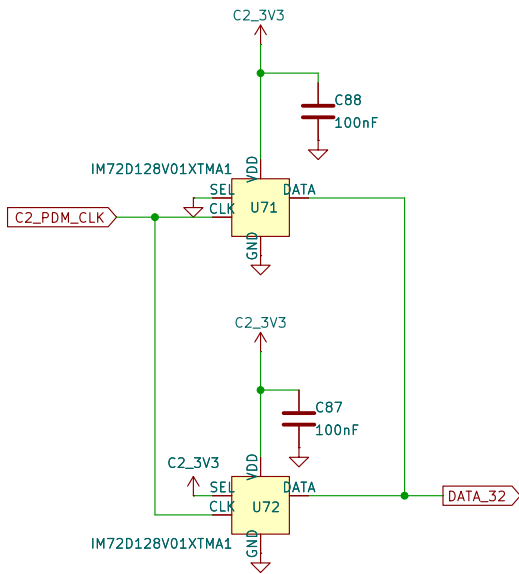
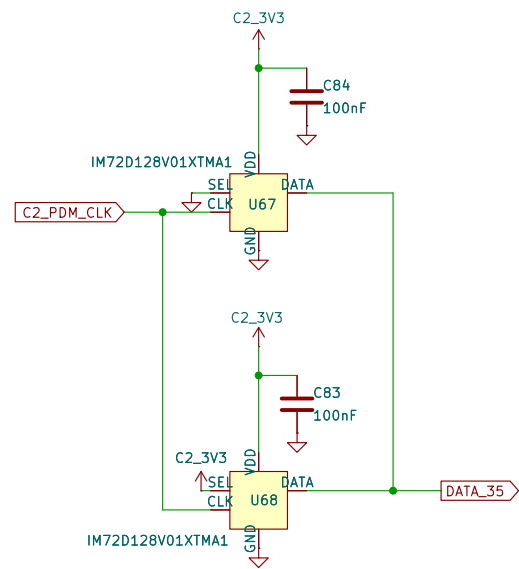
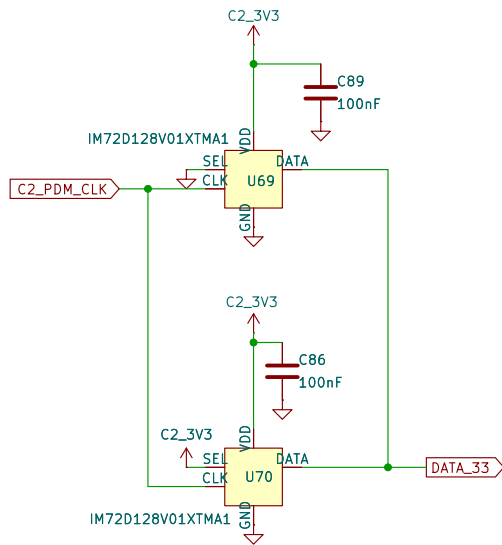
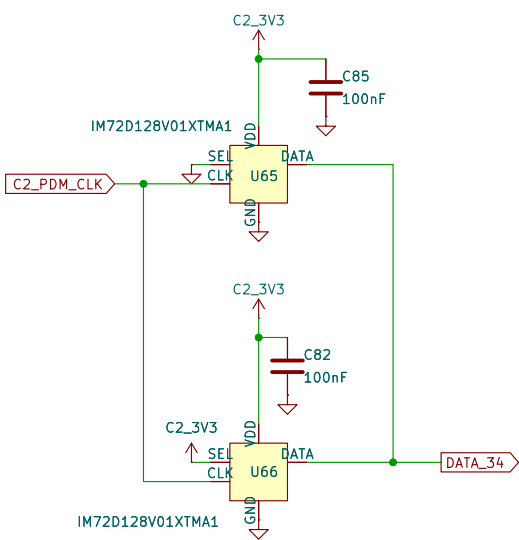
Title: 96-mic PDM Array – Arm 06 (mics 49-56)

Size: A3	Date:	Rev:
KiCad E.D.A. 10.0.5		Id: 13/18



SEL=GND → L channel (CLK falling); SEL=+3V3 → R channel (CLK rising)
VDD = +3V3; each mic has 100 nF local decoupling
IM72D128V01XTMA1 PDM MEMS mics

Sheet: /cluster_02/arm_07/ File: arm_07.kicad_sch		
Title: 96-mic PDM Array – Arm 07 (mics 57-64)		
Size: A3	Date:	Rev:
KiCad E.D.A. 10.0.5		Id: 14/18



SEL=GND → L channel (CLK falling); SEL=+3V3 → R channel (CLK rising)
VDD = +3V3; each mic has 100 nF local decoupling
IM72D128V01XTMA1 PDM MEMS mics

Sheet: /cluster_02/arm_08/ File: arm_08.kicad_sch		
Title: 96-mic PDM Array – Arm 08 (mics 65-72)		
Size: A3	Date:	Rev:
KiCad E.D.A. 10.0.5		Id: 15/18

arm_09



File: arm_09.kicad_sch

arm_10

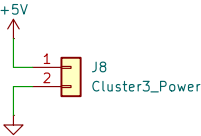
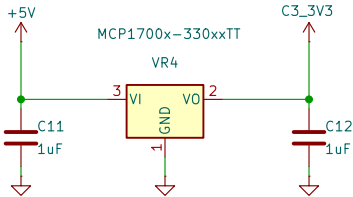
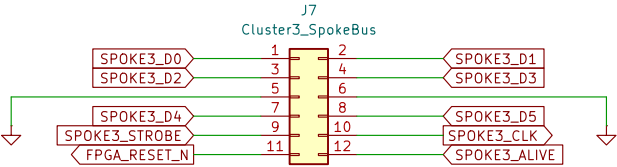
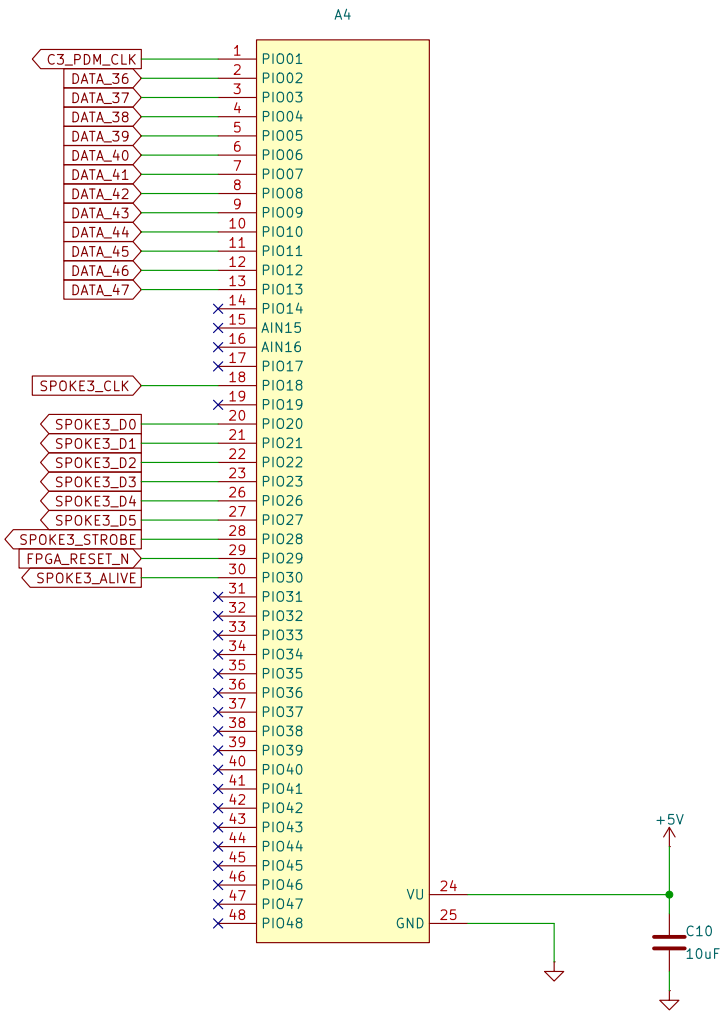


File: arm_10.kicad_sch

arm_11

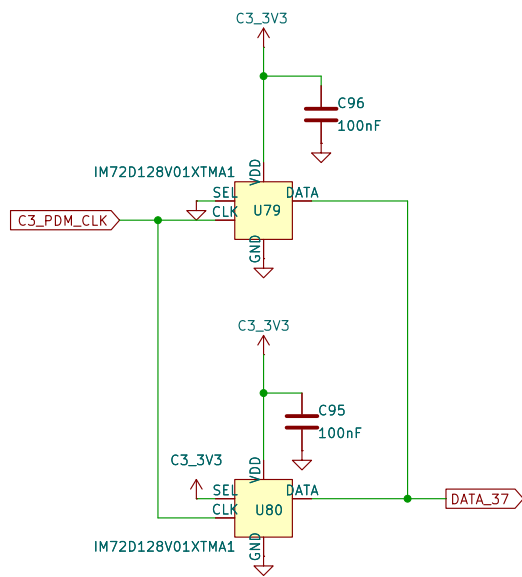
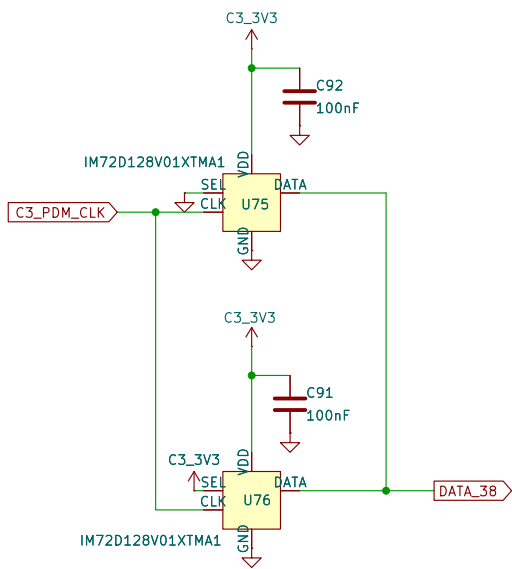
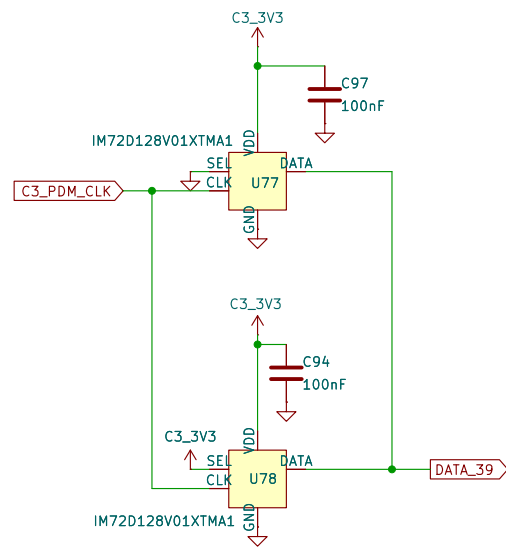
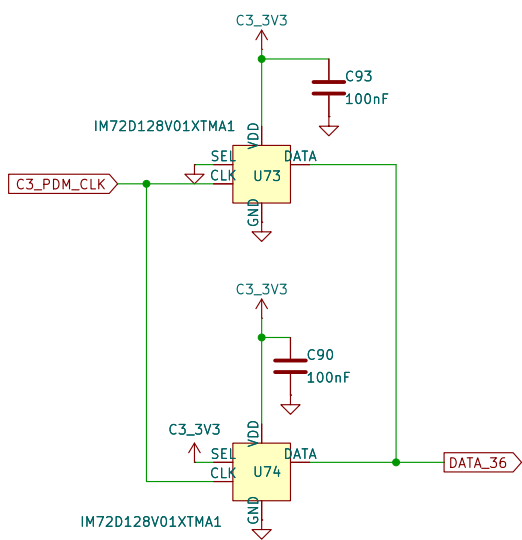


File: arm_11.kicad_sch



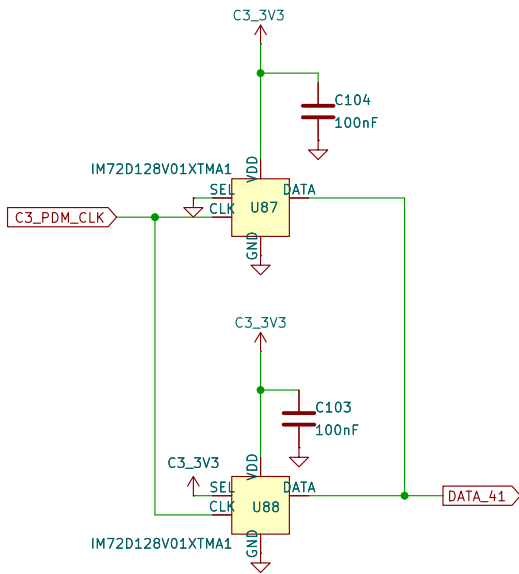
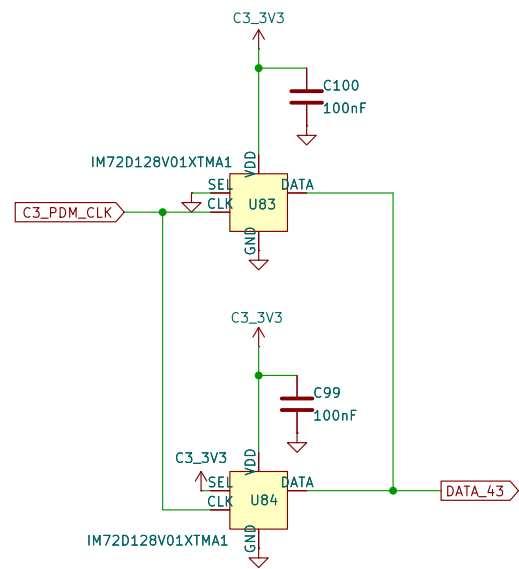
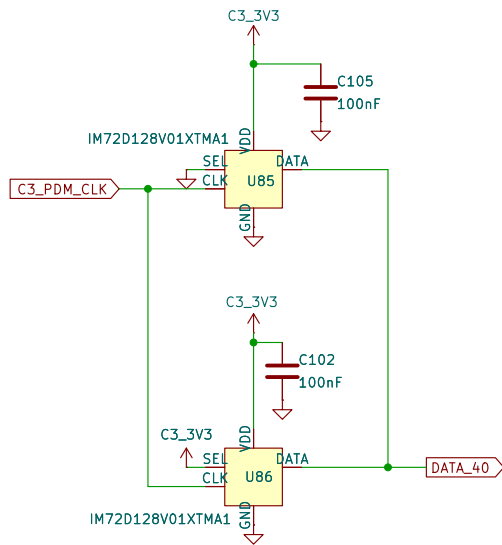
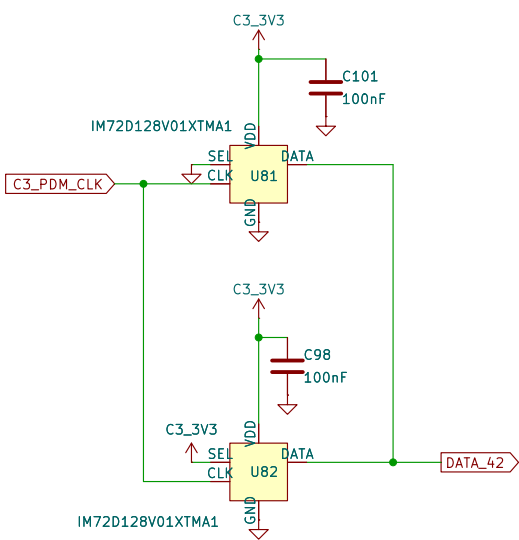
PDM: 12 data lines (2 mics/line) + 1 clock, to this cluster's 3 arms
Spoke bus: 6 data + strobe + fwd clock, parallel single-ended (not LVDS)
Digilent Cmod S7 (XC7S25) – CIC/FIR done in HDL, not shown here

Sheet: /cluster_03/ File: cluster_03.kicad_sch		
Title: Multi-FPGA Alternative – Cluster 3 (arms 9–11, mics 72–95)		
Size: A3	Date:	Rev:
KiCad E.D.A. 10.0.5		Id: 5/18



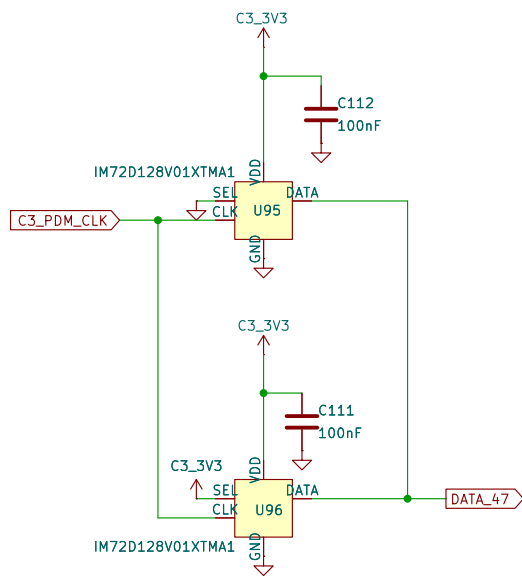
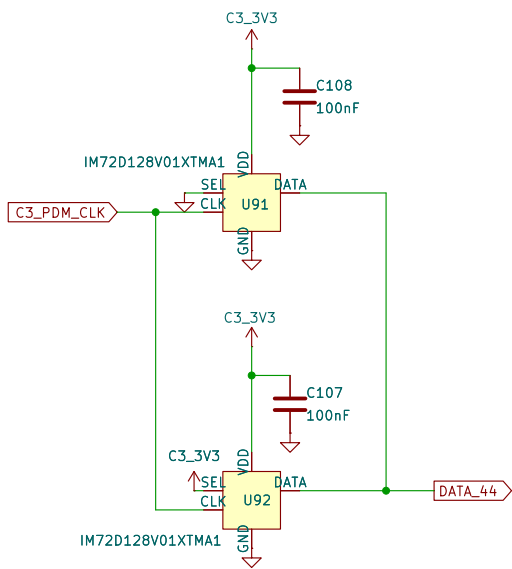
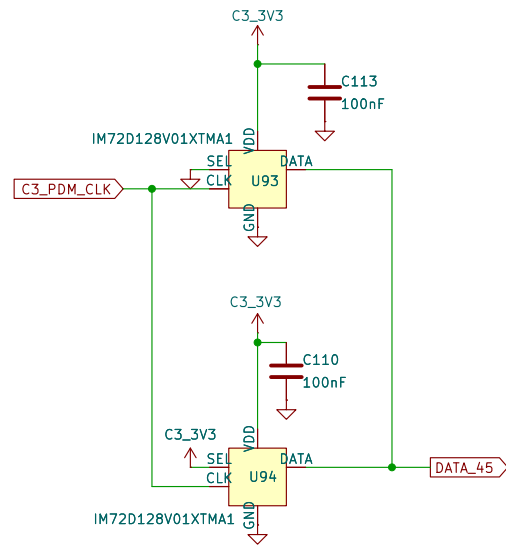
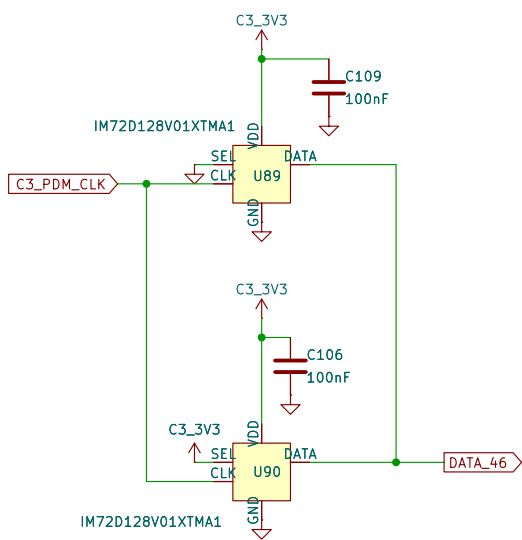
SEL=GND → L channel (CLK falling); SEL=+3V3 → R channel (CLK rising)
VDD = +3V3; each mic has 100 nF local decoupling
IM72D128V01XTMA1 PDM MEMS mics

Sheet: /cluster_03/arm_09/ File: arm_09.kicad_sch		
Title: 96-mic PDM Array – Arm 09 (mics 73-80)		
Size: A3	Date:	Rev:
KiCad E.D.A. 10.0.5		Id: 16/18



SEL=GND → L channel (CLK falling); SEL=+3V3 → R channel (CLK rising)
VDD = +3V3; each mic has 100 nF local decoupling
IM72D128V01XTMA1 PDM MEMS mics

Sheet: /cluster_03/arm_10/ File: arm_10.kicad_sch		
Title: 96-mic PDM Array – Arm 10 (mics 81-88)		
Size: A3	Date:	Rev:
KiCad E.D.A. 10.0.5		Id: 17/18

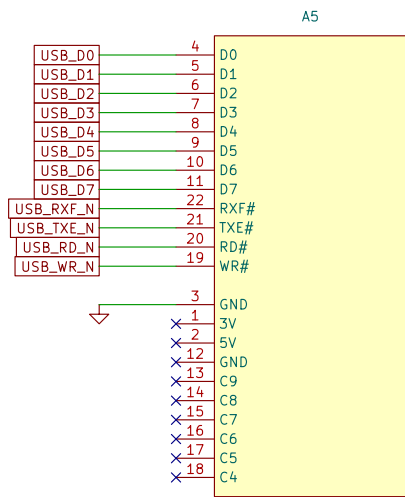
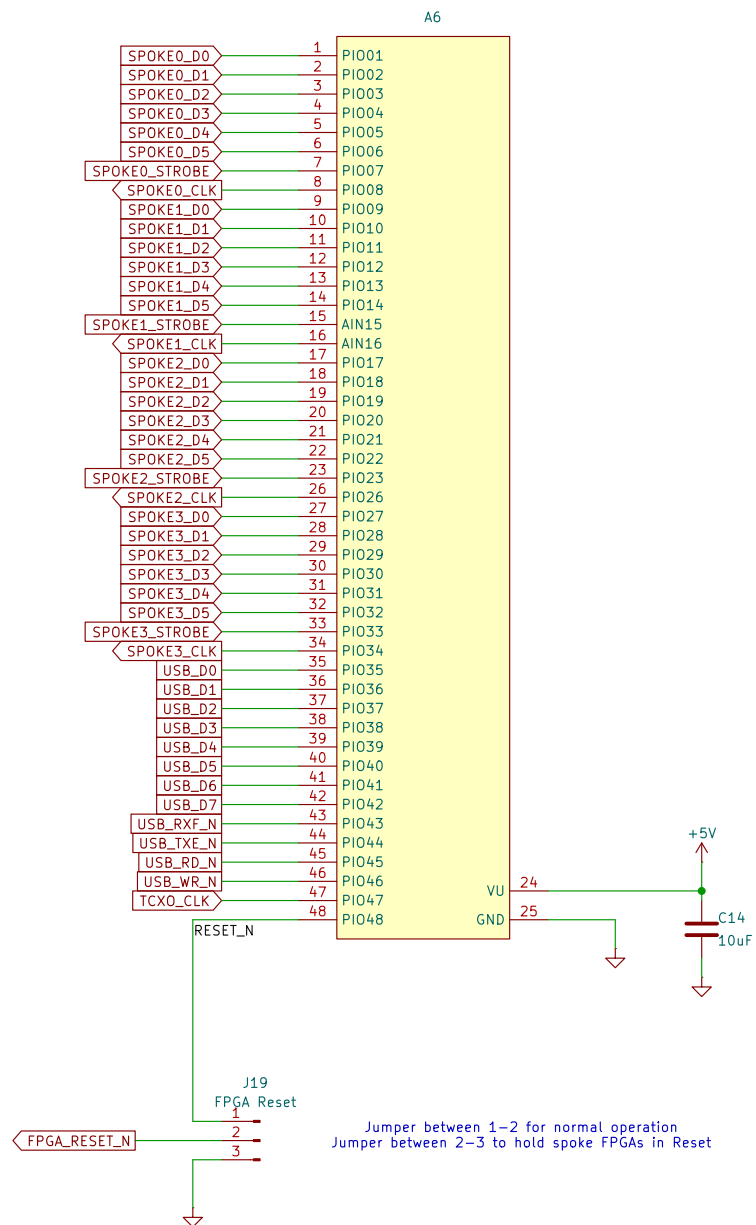
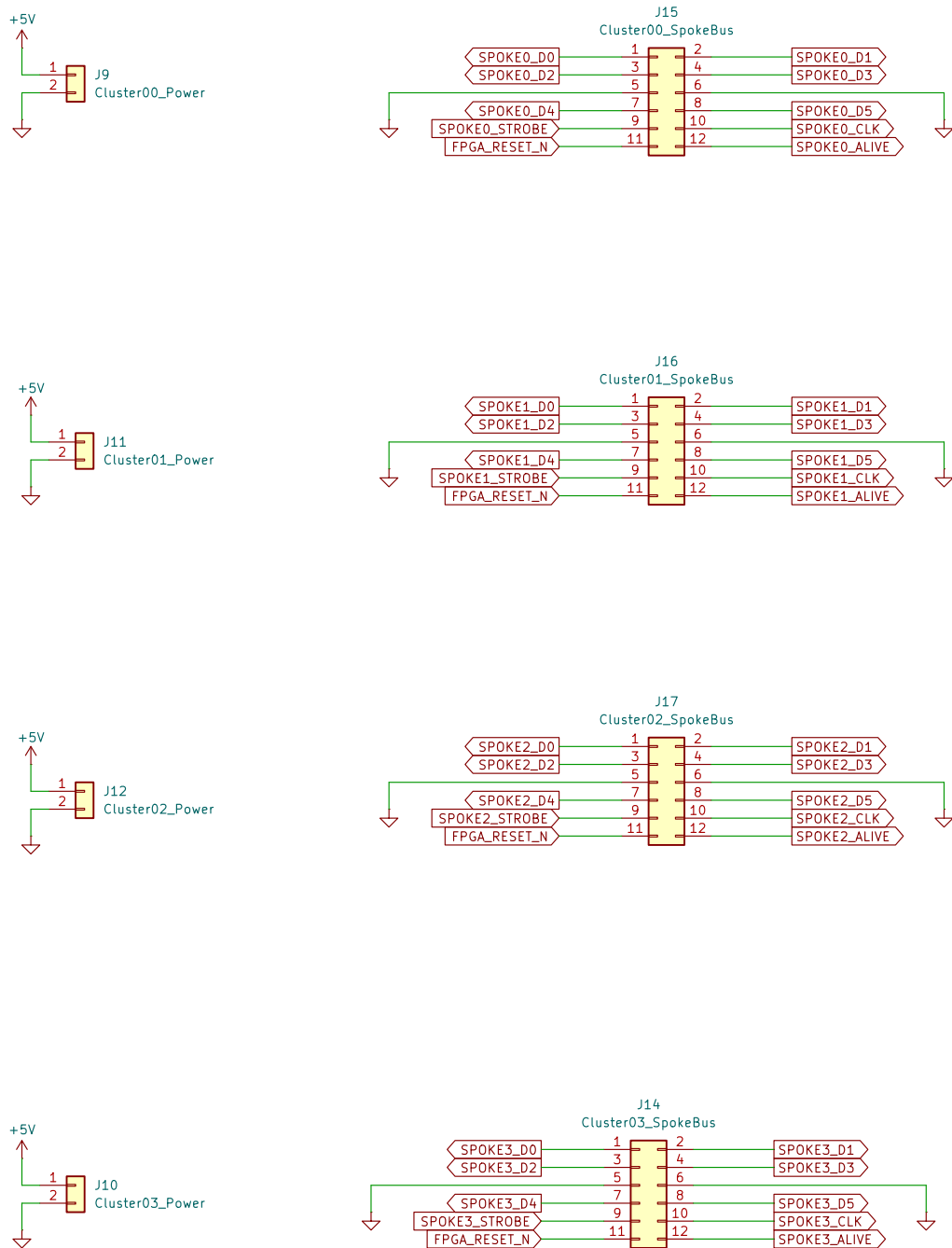


SEL=GND → L channel (CLK falling); SEL=+3V3 → R channel (CLK rising)
VDD = +3V3; each mic has 100 nF local decoupling
IM72D128V01XTMA1 PDM MEMS mics

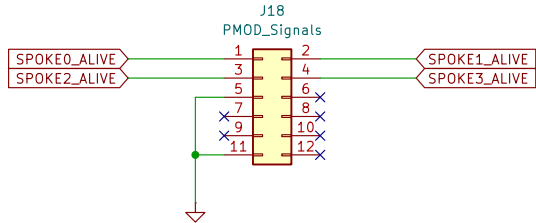
Sheet: /cluster_03/arm_11/
File: arm_11.kicad_sch

Title: 96-mic PDM Array – Arm 11 (mics 89-96)

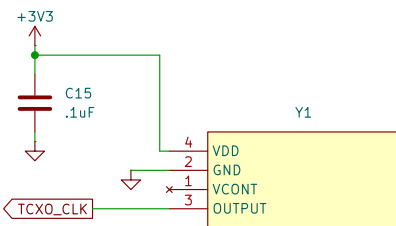
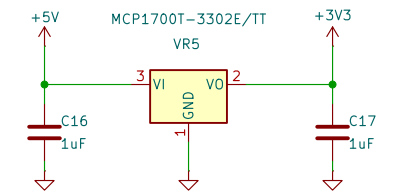
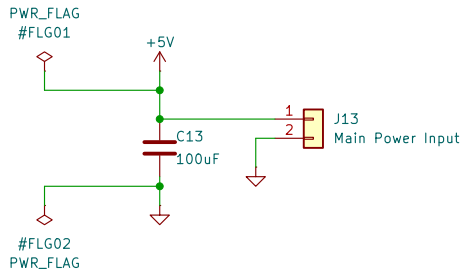
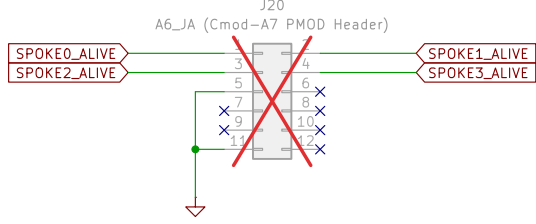
Size: A3	Date:	Rev:
KiCad E.D.A. 10.0.5		Id: 18/18



USB-C/Micro-B cable to Raspberry Pi 5, USB 3.0 port
Standalone: Pi 5 runs beamforming locally
Tethered: Pi 5 relays stream out its own on-board GbE to external host



Cable between J18 and the A6 PMOD connector



USB FIFO bridge (FT232H) to Raspberry Pi 5, also on DIP -- no GbE on hub
TCXO forwards 3.072 MHz PDM clock to all 4 clusters over spoke CLK
4x spoke bus (all on DIP header, no Pmod) -- reassembles 96ch